MATH.APP.220 Multivariable Calculus

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Contents (part 1)

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Function concepts

A function $f : \mathbb{R}^n \to \mathbb{R}^m$ is called a

- single variable real valued function, if $n = m = 1$
- single variable function, if $n = 1$
- multivariable function, if $n > 1$
- real valued function, if $m = 1$
- vector valued function, if $m > 1$

Single variable vector valued functions

- Consider the function $\mathbf{r} : [a, b] \to \mathbb{R}^n$,

$$\mathbf{r}(t) = (x_1(t), x_2(t), \ldots, x_n(t)).$$

  The functions $x_i(t)$ are called **coordinate functions** or **component functions** of the type $x_i : [a, b] \to \mathbb{R}$.

- In the cases $n = 2$ and $n = 3$ we often denote

$$\mathbf{r}(t) = x_1(t)\mathbf{i} + x_2(t)\mathbf{j}$$

$$\mathbf{r}(t) = x_1(t)\mathbf{i} + x_2(t)\mathbf{j} + x_3(t)\mathbf{k}.$$

Limits and continuity

- The **limit** of the function $\mathbf{r}(t)$ at $t_0 \in [a, b]$ is the vector

$$\lim_{t \to t_0} \mathbf{r}(t) = \left(\lim_{t \to t_0} x_1(t), \lim_{t \to t_0} x_2(t), \ldots, \lim_{t \to t_0} x_n(t) \right),$$

  if the limits of the coordinate functions exist.

- The function $\mathbf{r}(t)$ is **continuous at the point** t_0, if

$$\mathbf{r}(t_0) = \lim_{t \to t_0} \mathbf{r}(t)$$

  and **continuous**, if it is continuous at every point $t \in [a, b]$.

- The function $\mathbf{r}(t)$ is continuous, if and only if all of its component functions are continuous.

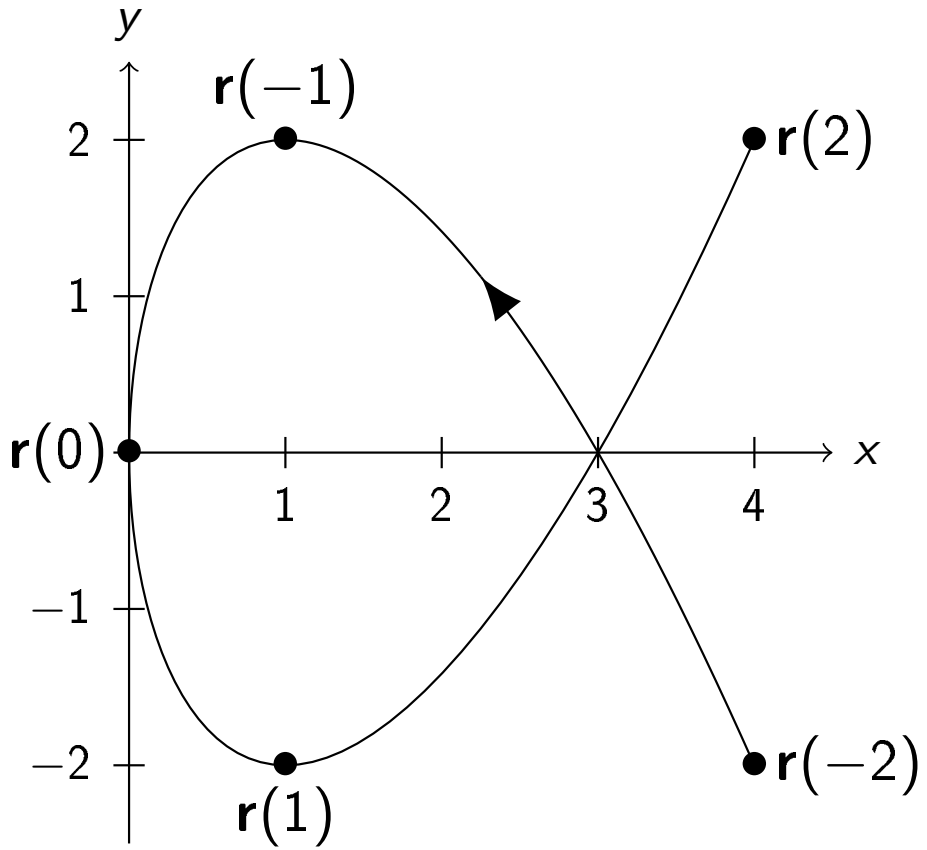
Definitions and concepts

- If a function $\mathbf{r} : [a, b] \to \mathbb{R}^n$ is continuous, it is called a **parametric curve**. The domain does not need to be a closed and bounded set.
- The image set $C = \mathbf{r}([a, b]) = \{\mathbf{r}(t) : t \in [a, b]\}$ is called a **curve**. If $n = 2$, the curve is a **plane curve** and if $n = 3$, a **space curve**.
- The point $\mathbf{r}(a)$ is the **initial point** and $\mathbf{r}(b)$ the **terminal point** of the curve.
- The (parametric) curve is **simple**, if it does not intersect itself, except possibly at its initial and terminal points.

Sketching a curve

Plot of the curve $\mathbf{r}(t) = \left(t^2, t^3 - 3t\right)$, $t \in [-2, 2]$.

t	$\mathbf{r}(t)$
-2	$(4, -2)$
$-3/2$	$(9/4, 9/8)$
-1	$(1, 2)$
$-1/2$	$(1/4, 11/8)$
0	$(0, 0)$
$1/2$	$(1/4, -11/8)$
1	$(1, -2)$
$3/2$	$(9/4, -9/8)$
2	$(4, 2)$



Parametrization

- If the graph $y = f(x)$ is represented in parametric form $\mathbf{r}(t)$, then $\mathbf{r}(t)$ is called the **parametrization** of f.
- We can parametrize any continuous function f with a so called **standard parametrization** by choosing the variable x as the parameter t, i.e.

$$\mathbf{r}(t) = (t, f(t)), \qquad t \in [a, b].$$

- It is often also possible to represent a parametric curve $\mathbf{r}(t)$ in the form $y = f(x)$ by a process called **elimination** of the parameter.

Parametrization of a line and a line segment

- You probably already know a way to parametrize a line by using (the position vector of) a point and a **direction vector**.
- If a line passes through the points $\mathbf{a}$ and $\mathbf{b}$, then $\mathbf{b}-\mathbf{a}$ is a direction vector for the line. In this case the direction is from $\mathbf{a}$ towards $\mathbf{b}$.
- The parametrization of a line is

$$\mathbf{r}(t) = \mathbf{a} + t(\mathbf{b}-\mathbf{a}), \qquad t \in \mathbb{R}.$$

- A line segment from the point $\mathbf{a}$ to the point $\mathbf{b}$ can be given with the parametrization

$$\mathbf{r}(t) = \mathbf{a} + t(\mathbf{b}-\mathbf{a}), \qquad t \in [0,1].$$

Several parametrizations for one curve

- The graph $y = f(x)$ of a function f has infinitely many parametrizations. For example, for $f(x) = x^2$, $x \in [-1, 1]$,

$$\mathbf{r}_1(t) = \left(t, t^2\right), \quad t \in [-1, 1],$$
$$\mathbf{r}_2(t) = \left(t-1, (t-1)^2\right), \quad t \in [0, 2].$$

- How to check, if two parametrizations give the same curve?

  $\rightarrow$ Eliminate the parameters from both parametrizations and also verify that they lead to the same interval of x.

- Keep in mind that even if two parametric curves have the same graph, the parametric curves are different if they have, for example, different initial points.

Some parametrizations of the unit circle

- $\mathbf{r}_1(t) = (\cos t, \sin t), \quad t \in [0, 2\pi]$
- $\mathbf{r}_2(t) = (\cos(2t), \sin(2t)), \quad t \in [0, \pi]$
- $\mathbf{r}_3(t) = (\cos(-t), \sin(-t)) = (\cos t, -\sin t), \quad t \in [0, 2\pi]$
- $\mathbf{r}_4(t) = \left(\cos\left(t + \frac{\pi}{2}\right), \sin\left(t + \frac{\pi}{2}\right)\right), \quad t \in [0, 2\pi]$
- $\mathbf{r}_5(t) = (\cos t, \sin t), \quad t \in [0, 5\pi]$
- $\mathbf{r}_6(t) = \left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2}\right), \quad t \in (-\infty, \infty)$

The derivative vector of a parametric curve

- $\mathbf{r} : [a, b] \to \mathbb{R}^n$ is **differentiable at the point** $t \in [a, b]$, if there exists
$$\mathbf{r}'(t) = \lim_{h \to 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}$$
and **differentiable**, if $\mathbf{r}$ is differentiable in every point $t \in [a, b]$. The limit $\mathbf{r}'(t)$ is a **derivative** (vector).
- A function $\mathbf{r}(t)$ is differentiable, if and only if all of its component functions are differentiable. Then
$$\mathbf{r}'(t) = \left(x_1'(t), x_2'(t), \ldots, x_n'(t)\right).$$

Interpretations of the derivative

- A geometric interpretation for the derivative: if $\mathbf{r}'(t) \neq \mathbf{0}$, then $\mathbf{r}'(t)$ is the tangent vector at the point $\mathbf{r}(t)$.
- In many applications, the parameter t represents time and $\mathbf{r}(t) = (x(t), y(t), z(t))$ is the location of an object (center of mass) at the instant t. Then the curve $C = \mathbf{r}([a,b])$ represents the route traversed by the object in the time interval $t \in [a,b]$. Then we define

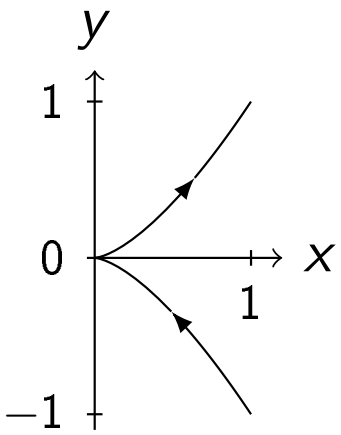
$$\mathbf{v}(t) = \mathbf{r}'(t) = \left(x'(t), y'(t), z'(t)\right) \qquad \text{velocity}$$

$$v(t) = \|\mathbf{v}(t)\| = \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2} \qquad \text{speed}$$

$$\mathbf{a}(t) = \mathbf{v}'(t) = \left(x''(t), y''(t), z''(t)\right) \qquad \text{acceleration}$$

Smoothness of a curve

- If a function $f : \mathbb{R} \to \mathbb{R}$ is differentiable, the graph $y = f(x)$ does not have **cusps** (sudden changes of direction).
- The parametric curve $\mathbf{r}(t) = \left(t^2, t^3\right)$ is differentiable, but its graph has a cusp. Differentiability does not imply smoothness.



- A parametric curve $\mathbf{r}$ is **smooth**, if $\mathbf{r}'(t)$ is continuous and $\mathbf{r}'(t) \neq \mathbf{0}$ for all $t \in [a, b]$.

On certain tangent vectors

- A tangent vector is horizontal, if its y coordinate is zero, but its x coordinate is not, in other words $x'(t) \neq 0$ and $y'(t) = 0$.
- Similarly, a tangent vector is vertical, if $x'(t) = 0$ and $y'(t) \neq 0$.
- Note that it is essential that only one of the component function derivatives is zero. If $\mathbf{r}'(t) = \mathbf{0}$, that is $x'(t) = 0$ and $y'(t) = 0$, the curve is not smooth, it has a cusp. At the cusp the curve has no tangent line.
- Do there exist such points (x, y), where a curve has at least two tangent vectors of separate directions?

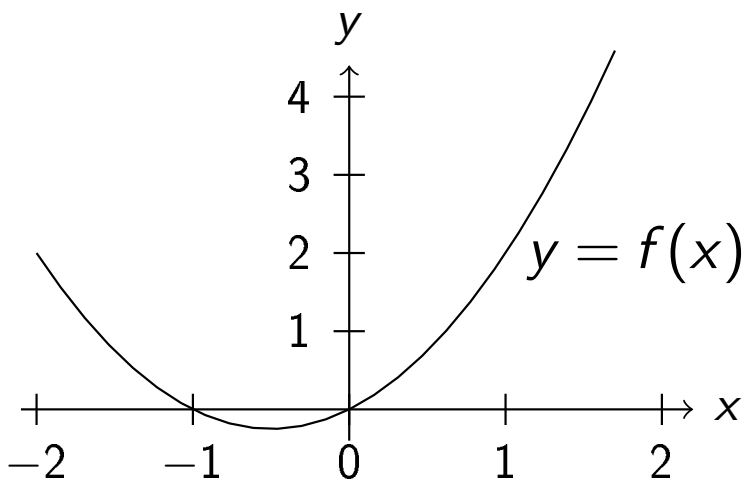
  $\rightarrow$ Yes, if the curve intersects itself, i.e. is not a simple curve.

Single variable real valued function

- The **graph** of a function $f : A \to \mathbb{R}$, where $A \subset \mathbb{R}$, is the two dimensional set

$$G_f = \left\{(x, y) \in \mathbb{R}^2 : x \in A, y = f(x) \in \mathbb{R}\right\}$$

that you can plot in the xy-plane.

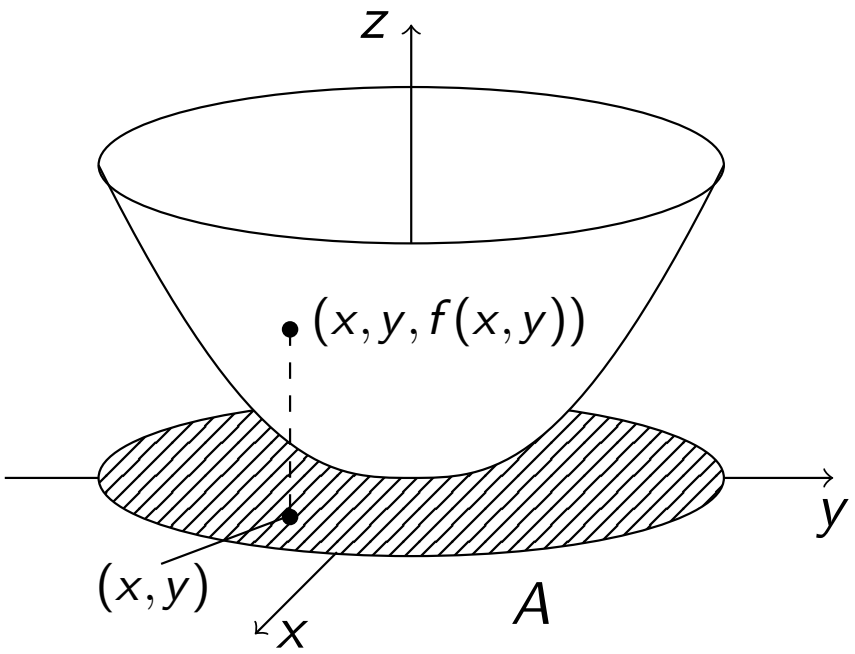


Real valued function of two variables

- The graph of a function $f : A \to \mathbb{R}$, where $A \subset \mathbb{R}^2$, is the set

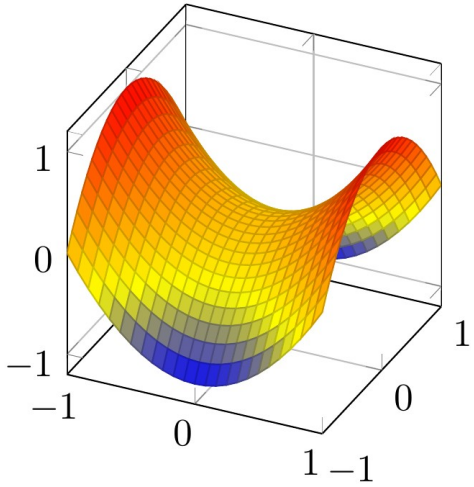
$$G_f = \left\{(x, y, z) \in \mathbb{R}^3 : (x, y) \in A, z = f(x, y) \in \mathbb{R}\right\},$$

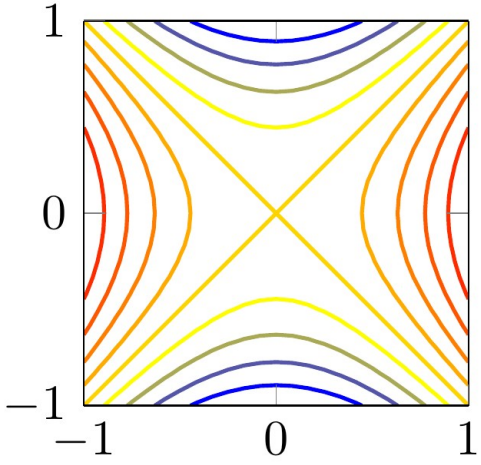
which is a surface that you can plot in the xyz-space.



Level curves

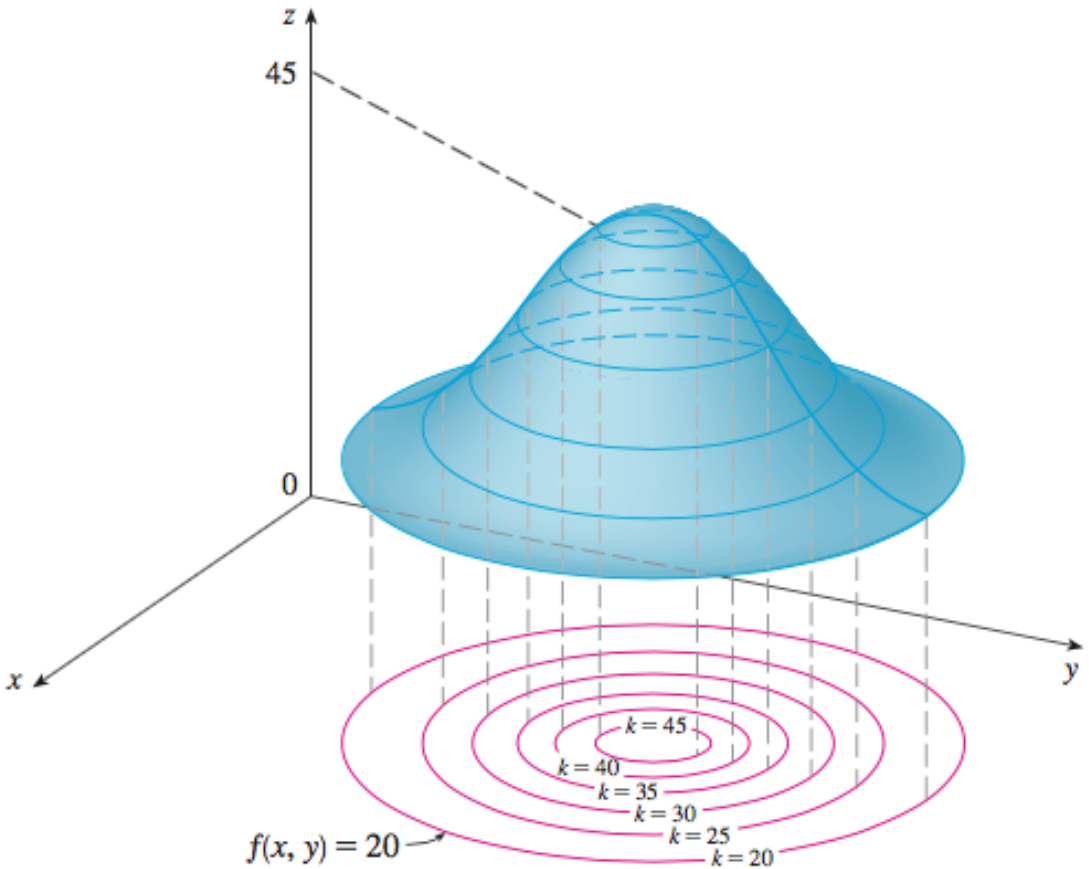
- A function of type $\mathbb{R}^2 \to \mathbb{R}$ can also be visualized by plotting **level curves**. A level curve is a set of points (x, y) for which $f(x, y) = c$, where c is a constant.
- Depicted below is the graph of $f(x, y) = x^2 - y^2$ and its level curves. For $c = 0$, i.e. when $x^2 - y^2 = 0$, we get the lines $y = x$ and $y = -x$. Level curves for other values of c are hyperbolas.





Level curves vs. contours

A level curve is an intersection curve of the function graph (the surface) and a horizontal plane, projected on the xy-plane.



Representations of certain surfaces

- In the following functions $f(x)$, replacing the variable x with the expression $\sqrt{x^2+y^2}$ creates a function $g(x,y)$. The graph $z=g(x,y)$ is a **surface of revolution** obtained by rotating the plot of $z=f(x)$ of the xz-plane about the z-axis.
- $f(x)=x \Longrightarrow g(x,y)=\sqrt{x^2+y^2}$
  (rotating a line gives us a cone)
- $f(x)=x^2 \Longrightarrow g(x,y)=\left(\sqrt{x^2+y^2}\right)^2=x^2+y^2$
  (parabola $\rightarrow$ paraboloid)
- $f(x)=\sqrt{a^2-x^2} \Longrightarrow g(x,y)=\sqrt{a^2-x^2-y^2}$
  (semicircle $\rightarrow$ hemisphere, i.e. a half sphere)